AdaptRAG: Adaptive Retrieval-Augmented Generation for Reducing Hallucinations in Large Language Models

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Abstract

Large language models (LLMs) generate fluent but often factually incorrect text, a phenomenon known as hallucination that limits their reliability in knowledge-critical applications. Retrieval-augmented generation (RAG) mitigates this by grounding output in retrieved evidence, yet current approaches apply fixed retrieval strategies regardless of query complexity — under-retrieving for complex multi-hop questions and over-retrieving noisy context for simple factual queries. We propose **AdaptRAG**, an adaptive retrieval-augmented generation framework that dynamically adjusts retrieval depth, source selection, and context integration based on query complexity. AdaptRAG employs a lightweight query complexity classifier that categorizes queries into three tiers (simple, intermediate, complex), each triggering a distinct retrieval strategy ranging from shallow single-source lookup to deep multi-source iterative retrieval with cross-encoder re-ranking. We evaluate AdaptRAG on three hallucination benchmarks — TruthfulQA, HaluEval, and FActScore — using LLaMA-3-8B and Mistral-7B as backbones. AdaptRAG achieves 12–18% reduction in hallucination rate compared to the strongest baseline (Self-RAG), with the largest gains on complex multi-hop queries (+9.7 points). Ablation analysis confirms that adaptive retrieval depth is the single most impactful component, while the classifier adds only 30ms overhead per query.

1 Introduction

Large language models have demonstrated remarkable capabilities across a wide range of natural language tasks [Brown et al., 2020, Touvron et al., 2023]. However, their tendency to generate plausible-sounding but factually incorrect content — hallucination — poses a significant barrier to deployment in high-stakes domains [Huang et al., 2023]. Recent studies estimate that even state-of-the-art models hallucinate on 15–25% of factual queries [Mallen et al., 2023, Min et al., 2023].

Retrieval-augmented generation (RAG) addresses this by grounding output in retrieved evidence [Lewis et al., 2020]. Despite its success, current implementations apply **fixed retrieval strategies** regardless of query complexity, leading to under-retrieval for complex queries and over-retrieval for simple ones [Yoran et al., 2024].

Recent adaptive approaches make binary retrieve/don't-retrieve decisions [Asai et al., 2024, Jiang et al., 2023, Yan et al., 2024] but do not adapt *how much* to retrieve or *from where*. We propose **AdaptRAG** with four contributions:

- A **query complexity classifier** (87% accuracy, 30ms overhead) that categorizes queries into three complexity tiers.
- A **3-tier adaptive retrieval strategy** adjusting depth (k=2, 5, or 10), re-ranking, and iteration.

- **Multi-source routing** directing retrieval to appropriate knowledge sources.
- A **confidence feedback loop** triggering re-retrieval for low-confidence claims.

2 Related Work

Retrieval-Augmented Generation. The RAG paradigm [Lewis et al., 2020] combines retrieval with generation. RETRO [Borgeaud et al., 2022] and FiD [Izacard and Grave, 2021] extended it at scale. All use fixed retrieval depth.

Adaptive Retrieval. Self-RAG [Asai et al., 2024] uses reflection tokens; FLARE [Jiang et al., 2023] uses token confidence; CRAG [Yan et al., 2024] adds corrective evaluation; RePlug [Shi et al., 2024] uses ensemble reweighting. None adapts retrieval depth or source.

Hallucination Evaluation. TruthfulQA [Lin et al., 2022], HaluEval [Li et al., 2023], and FActScore [Min et al., 2023] provide benchmarks. Existing mitigation is reactive; our approach is proactive.

3 Methodology

3.1 Query Complexity Classifier

We fine-tune DeBERTa-v3-base with a 3-class head (Simple, Intermediate, Complex) on 10K queries from NQ, TriviaQA, HotpotQA, SQuAD 2.0, and StrategyQA. Accuracy: 87.3%.

3.2 Adaptive Retrieval Strategy

- **Tier 1 (Simple):** k=2, BM25+Contriever, no re-ranking.
- **Tier 2 (Intermediate):** k=5, hybrid + cross-encoder re-ranking.
- **Tier 3 (Complex):** k=10, multi-source + iterative (2 rounds).

3.3 Context-Aware Generator

Retrieved passages are formatted in a structured prompt. A confidence feedback loop (threshold τ=0.3) triggers re-retrieval for ∼15–20% of responses.

4 Experiments

4.1 Setup

Benchmarks: TruthfulQA (MC1), HaluEval (F1), FActScore. Baselines: No Retrieval, Naive RAG, Self-RAG, FLARE, CRAG, RePlug. Backbones: LLaMA-3-8B, Mistral-7B. Corpus: Wikipedia Dec 2024 (∼33M passages).

4.2 Main Results

AdaptRAG outperforms Self-RAG by 6.3/6.6/6.6 points across all benchmarks.

Table 1: Main results (%). Best in **bold**. LLaMA-3-8B backbone.

Method	TruthfulQA	HaluEval	FActScore
No Retrieval	34.2	62.1	58.3
Naive RAG	41.7	68.4	65.1
RePlug	43.5	69.7	67.2
FLARE	45.9	70.8	68.7
CRAG	47.1	72.5	70.3
Self-RAG	48.3	73.2	71.8
AdaptRAG	**54.6**	**79.8**	**78.4**

4.3 Ablation Study

Table 2: Ablation on TruthfulQA MC1.

Configuration	MC1	Δ
Full AdaptRAG	54.6	—
− Adaptive depth	48.1	−6.5
− Query classifier	49.8	−4.8
− Multi-source routing	51.2	−3.4
− Confidence feedback	52.1	−2.5

4.4 Per-Complexity Analysis

Table 3: TruthfulQA MC1 by query complexity tier.

Tier	#	AdaptRAG	Self-RAG	Δ
Simple	312	72.1	68.4	+3.7
Intermediate	298	51.3	44.2	+7.1
Complex	207	38.6	28.9	**+9.7**

Gains scale with complexity: +3.7 (Simple), +7.1 (Intermediate), +9.7 (Complex).

4.5 Efficiency

AdaptRAG retrieves 4.2 docs/query on average (vs. 5.0 for Naive RAG) with 340ms latency (9.7% overhead).

4.6 Limitations

English-only evaluation; 87% classifier accuracy; web search API dependency; factual/extractive tasks only.

5 Conclusion

We presented AdaptRAG, achieving 12–18% hallucination reduction by adapting retrieval to query complexity. A lightweight classifier (30ms overhead) drives meaningful adaptation, with the largest gains on complex queries (+9.7 points). Future work includes multilingual extension, continuous depth learning, and integration with training-time methods.

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